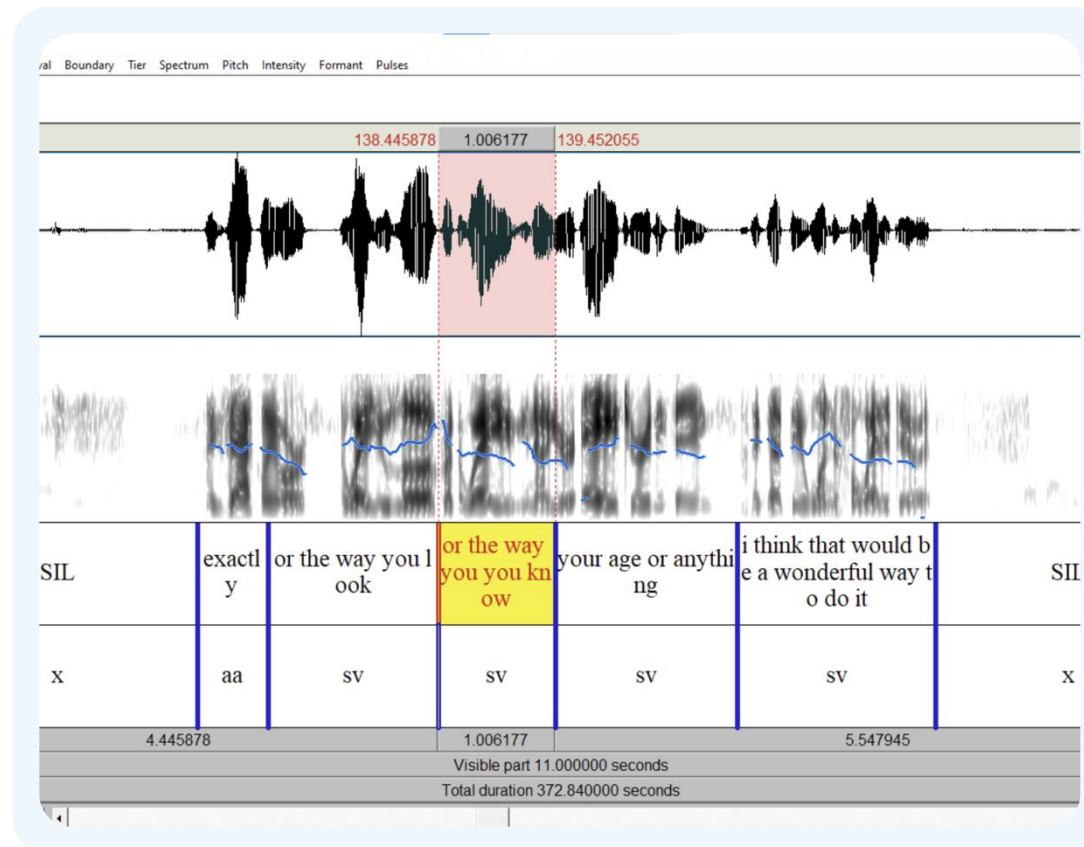


Disentangling Annotator Skill from Verifier Strictness

in Cross-Verified Dialogue Annotation

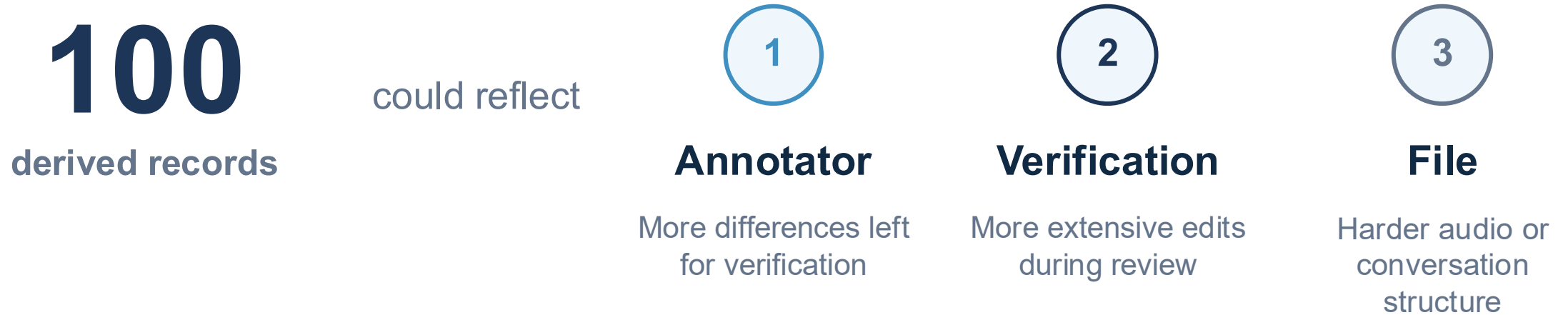
Zihao Tao · John A. Prado · Ignazio Steven LaManna
Ryan Puterbaugh · Mim Datta · Julia Hirschberg

Columbia University · University of Alberta



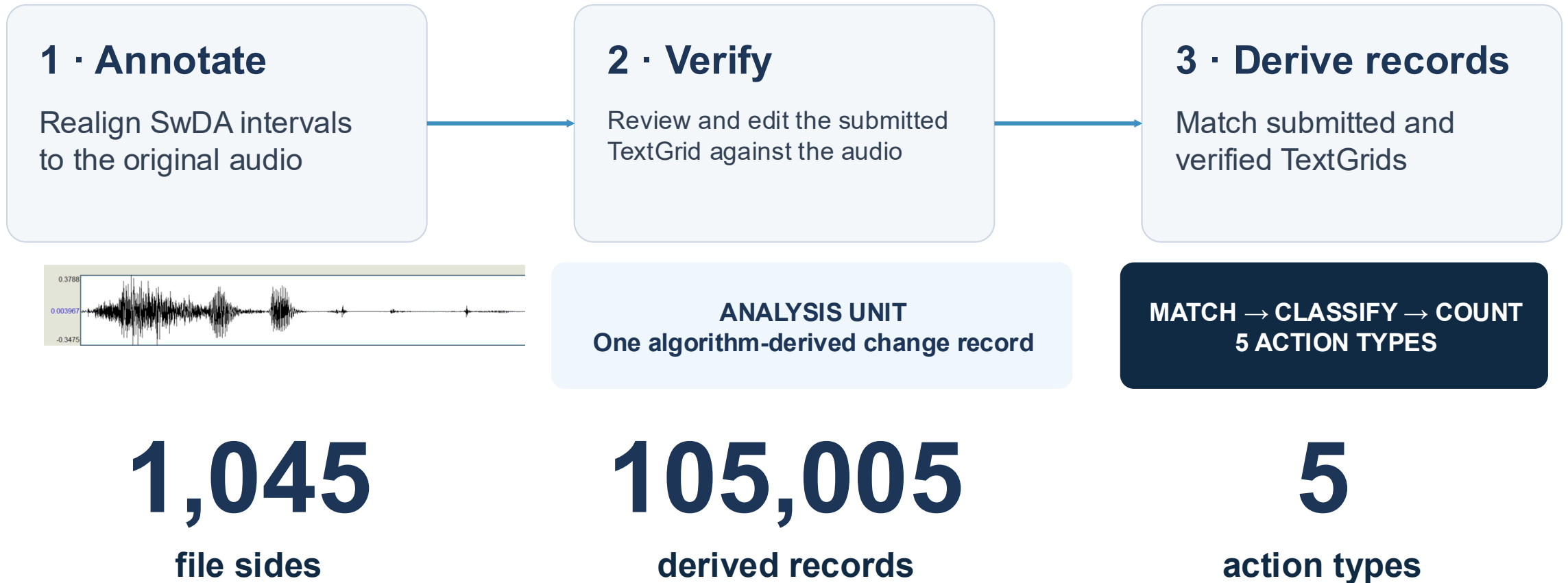
What does the paper's correction count actually measure?

One observed count can have three workflow sources



Observed record count $\neq$ annotator quality

RASwDA derives analysis records from two TextGrid versions



Crossed assignments identify role effects

Verifier

Annot.	T1	T2	T3	T4	T5
T1	—	164	63	13	43
T2	49	—	44	16	13
T3	38	38	—	46	10

Every annotator

was checked by all four other team members

537

complete cases
with both roles
identified

Off-diagonal overlap separates annotator and verifier effects.

The model asks two symmetric questions

Outcome used in the paper

normalized correction rate = derived change records / non-silence intervals

Do annotators differ

after controlling for
who verified the file?

Do verifiers differ

after controlling for
who annotated the file?

Controls personal batch index · verifier batch index · log(file length)

Random effects annotator · verifier · batch within annotator · file residual

Residual dominates; verifier variance = 0

10.8%

Annotator identity

modest component

0.0%

Verifier identity

singular-fit boundary

80.6%

File-level residual

difficulty + measurement noise

8.7%

batch within annotator



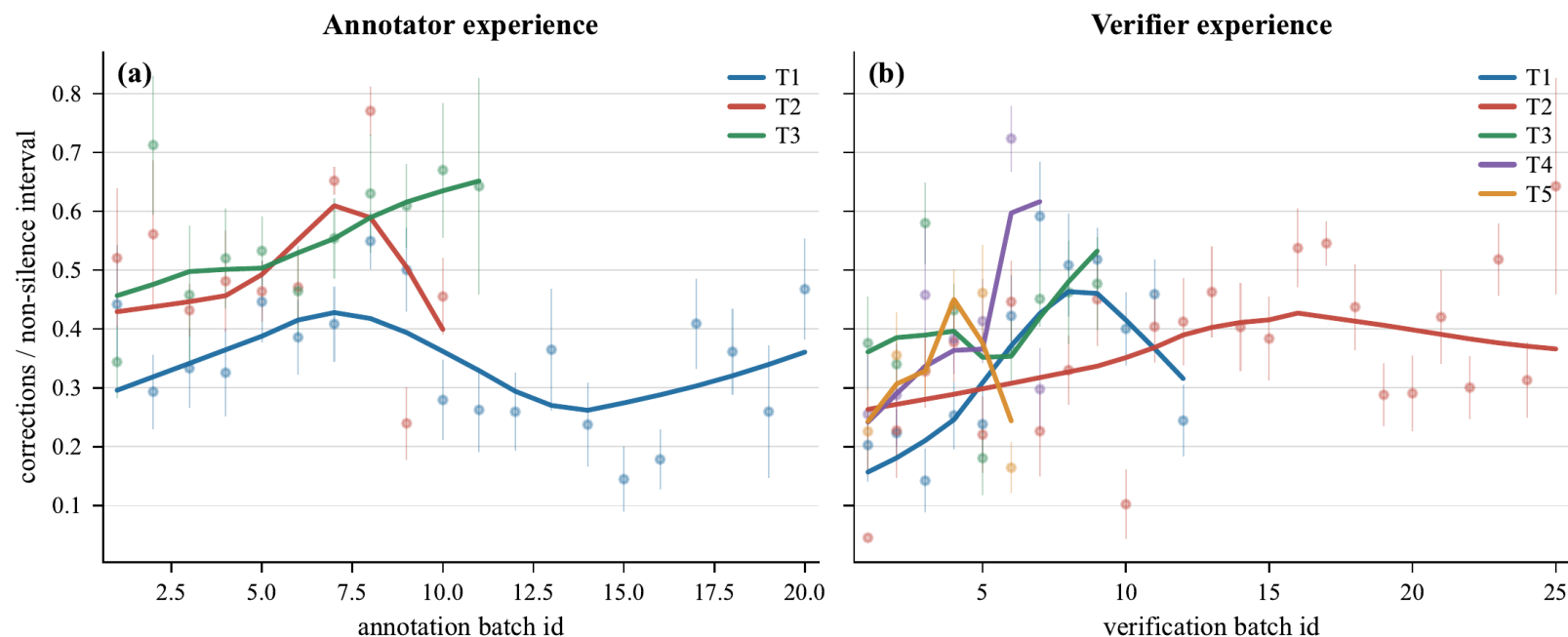
10.8 annotator

8.7 batch

80.6 file residual

No detectable verifier effect ≠ no verification value

Experience does not explain the remaining variation



Annotator batch

$p = 0.60$

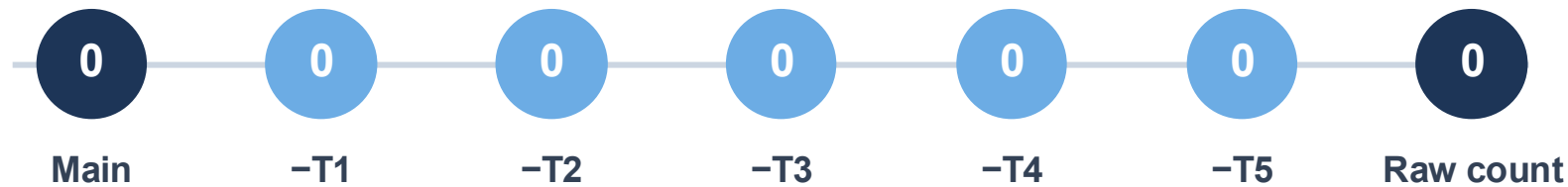
Verifier batch

$p = 0.06$

No coherent
downward drift

Estimated verifier variance = 0 in all seven fits

$\sigma^2_v = 0$ in all seven fits



Power limit

5 verifiers
537 complete cases
Small effects may remain

No detectable effect $\neq$ identical verifiers

Most reconstructed changes involve boundary placement

Share of 105,005 derived change records



58.5%

audio-anchored
boundary shifts

3.0%

pure dialogue-act
label changes

Boundary-shift records are audio-anchored, not pure label judgments.

Audio anchors boundaries; label shares vary



Audio constrains where speech begins and ends

Semantic choices leave room to diverge

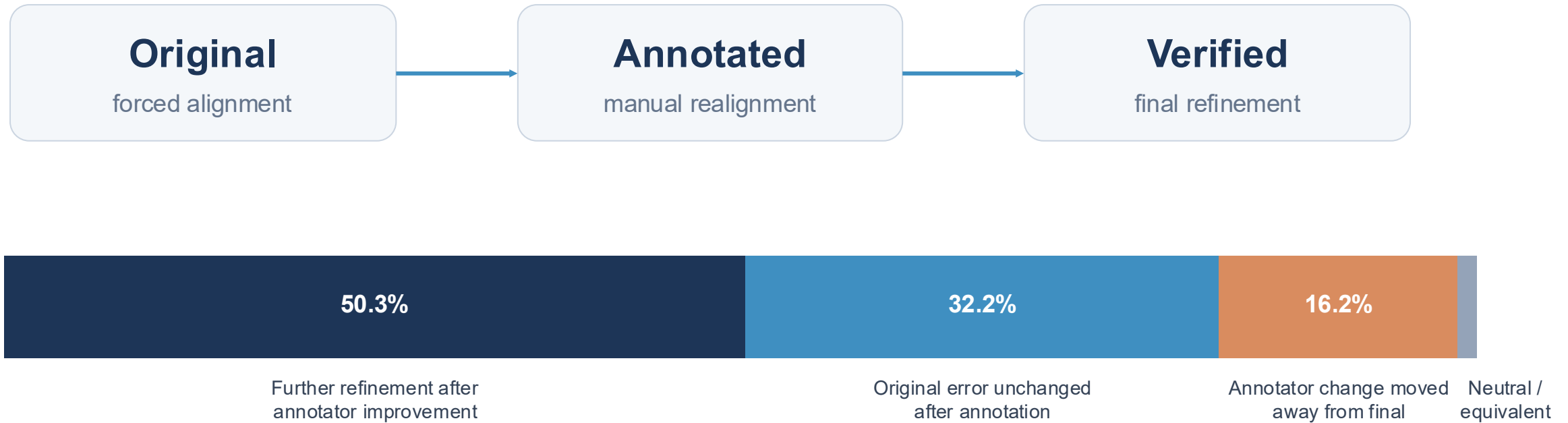
sd ↔ sv b / aa
/ ny

Semantic-edit share



T3: 12.8% DA_CHANGE + 7.5% MODIFY

Version differences are not always annotator errors



The derived rate is a process measure—not a pure error rate.

Make verification auditable

1

Preserve evidence

Log edit events directly—or derive typed records from versioned files—with metadata.

2

Cross assignments

Use at least two verifiers per annotator, with overlap.

3

Release data + method

Share versions, derived records, and comparison code.

For RASwDA: improve boundary-placement support first.

Correction count is a derived joint measure

- **Crossed design + derived records**

separate annotator, verifier, batch, and residual components.

- **10.8% vs. 0.0%**

Annotator identity matters modestly; verifier identity does not detectably differ.

- **58.5% boundary-shift records**

Audio anchoring is consistent with limited room for verifier-specific differences.



Read the paper

aclanthology.org/2026.sigdial-1.29

Questions?